

CHAPTER 4

DEEP LEARNING MODELS FOR WATER QUALITY PREDICTION IN AQUACULTURE

This chapter delves into applying DL models, such as LSTM, GRU, and their bidirectional counterparts, for WQP in aquaculture. The primary focus of this study lies in comparing these DL models in terms of accuracy. DL models can consider the non-linear characteristics of WQ changes, enabling them to capture intricate patterns in the data. This unique trait allows DL models to identify essential features that efficiently provide concise and reliable results [81]. Among the DL models, LSTM and GRU are widely used for time series prediction due to their remarkable performance [82]. In this chapter, Section 4.1 provides an insightful introduction, highlighting the significant contributions of this work. Section 4.2 of this chapter comprehensively explains the ADAK dataset, offering detailed insights into the data characteristics and pre-processing. Moving on to Section 4.3, the DL models employed for WQP in this study are explained. The section outlines the architecture and underlying mechanisms of each DL-RNN model, focusing on LSTM, GRU, and their bidirectional counterparts (Bi-LSTM and Bi-GRU). These DL models have been selected for their proficiency in capturing temporal patterns and sequences, making them suitable candidates for WQ time series prediction. Section 4.4 is dedicated to result analysis, a critical component of this study focusing on comparing the performances of the different DL models, including LSTM, GRU, Bi-LSTM, and Bi-GRU. The chapter is summarized in Section 4.5.

4.1 INTRODUCTION

A WQP model could describe the relationship between monitored data as well as trends in WQ parameters based on available data gathered by the WQM system and forecast changes in WQ for a specified period. There are many shortcomings in conventional prediction approaches, such as weak forecasting precision, least stability etc. Novel prediction methods focused on artificial intelligence have emerged with the development of computational intelligence and bionic technology. Traditional methods for estimating WQ mainly include: the Markov model based on fuzzy-set theory [66, 67], the Grey Markov chain model method [68, 69], the method of prediction of regression [70, 71], the method of time series [72, 73], methods of WQP

model [74, 75], the ANN method [76, 77], the LS-SVM prediction method [78, 79], the combination prediction method [80]. These approaches offer an efficient method for forecasting WQ but are not suitable for aquaculture. WQ is affected by many factors such as complex inter-relationship between the parameters, non linearity in the WQ changes etc. Hence these models results in poor prediction accuracy.

LSTM was introduced to enhance long sequence learning and avoid the vanishing-gradient issue of RNN [138]. GRU DL-RNN was proposed in [139] and it has performance comparable with LSTM in terms of accuracy and lower demand for processing resources. Some reported research works based on LSTM model resolves the issues with conventional prediction methods. The authors have used LSTM to predict drinking WQ in [85], and in [86] for mariculture WQP. In order to further improve the performance Adam an efficient stochastic optimisation method is used [88, 89]. Adam combines advantage of AdaGrad [90] and RMSProp [91], which are two popular methods.

WQP is a typical time series problem, which can be considered to be a sequential regression problem. For training the model, this sequential time series data is converted into a set of feature values and the corresponding target value using the sliding window procedure since the regression is a supervised learning problem. The predicted output depends on the current input and the entire length of the time sequence with different levels of importance or weightage. Many prediction models map relationships with the input and output data at a particular time, ignoring the features in the lengthy sequence. To capture the features from this whole sequence data, different DL models can be used in the DL layer. Bidirectional DL models can be used for the utilization of the information from the sequence data in the forward and backward directions. In the bidirectional NNs, with their non-linear mapping of the sequence, the DL layer accurately captures the sequence correlation of data to provide more precise prediction results. LSTM and GRU are two types of DL-RNN models used for sequence modeling tasks.

The significant contributions of the work presented in this chapter can be highlighted as follows:

1. This work conducts a comprehensive evaluation of standalone DL models, including LSTM, GRU, Bi-LSTM, and Bi-GRU, for accurate WQP. The models are trained and tested on diverse WQ parameters and datasets, providing a comprehensive understanding of their performance.
2. An in-depth analysis of the models' performance is carried out, considering metrics such as MAE, MSE, RMSE, and MAPE. The analysis reveals the strengths

and limitations of each model in predicting different WQ parameters.

4.2 DATA COLLECTION AND ANALYSIS

4.2.1 Acquisition of data

The data of aquaculture WQ parameters used in this work is collected from an aquaculture farm under ADAK at Kollam, Kerala, India. The WQ parameter data for three years, from January 2016 to December 2018, is collected on a daily basis from the pond. The trends in variations of WQ parameters of the aquaculture pond are studied. Figure 4.1 shows the annual trends of salinity, pH, DO, and temperature. From the plots, it is clear that all WQ parameters are non-stationary in nature. Table 4.1 shows

Table 4.1: Average water quality parameters monitored in 2016, 2017 and 2018

Water quality parameter	Year	Mean	Min	Max
Salinity (ppt)	2016	14.3791	13.186	15.736
pH		7.3338	7.192	7.458
DO (ml/L)		5.3581	5.237	5.512
Temperature (°C)		24.6277	23.299	25.685
Salinity (ppt)	2017	15.4084	14.261	17.395
pH		7.3326	7.192	7.46
DO (ml/L)		5.2995	5.058	5.499
Temperature (°C)		24.9244	22.999	27.278
Salinity (ppt)	2018	15.2589	14.307	16.818
pH		7.3411	7.24	7.483
DO (ml/L)		5.2856	5.116	5.391
Temperature (°C)		25.1092	24.294	26.681

the average of the WQ parameter data of each year. The salinity in 2016 was between 13.18 ppt and 15.73 ppt; in 2017, it was between 14.26 ppt and 17.39 ppt, and in 2018 it was between 14.30 ppt and 16.82 ppt. The pH in 2016 was between 7.20 and 7.45, and in 2017, it was between 7.19 and 7.46. Moreover, in 2018, it was between 7.24 and 7.48. The DO in 2016 was between 5.23 ml/L and 5.51 ml/L; in 2017, it was between 5.05 ml/L and 5.49 ml/L, and in 2018 it was between 5.11 ml/L and 5.39 ml/L. The temperature value in 2016 was between 23.30 °C and 25.69 °C; in 2017 it was between 22.99 °C and 27.27 °C, and in 2018 it was between 24.29 °C and 26.68 °C. The mean DO minimum in 2018, where the mean temperature is also highest, indicates the relation between DO and temperature. When temperature increases, DO will decrease. In all three years, the average salinity value is coming down in June - July, coinciding with the monsoon season in Kerala.

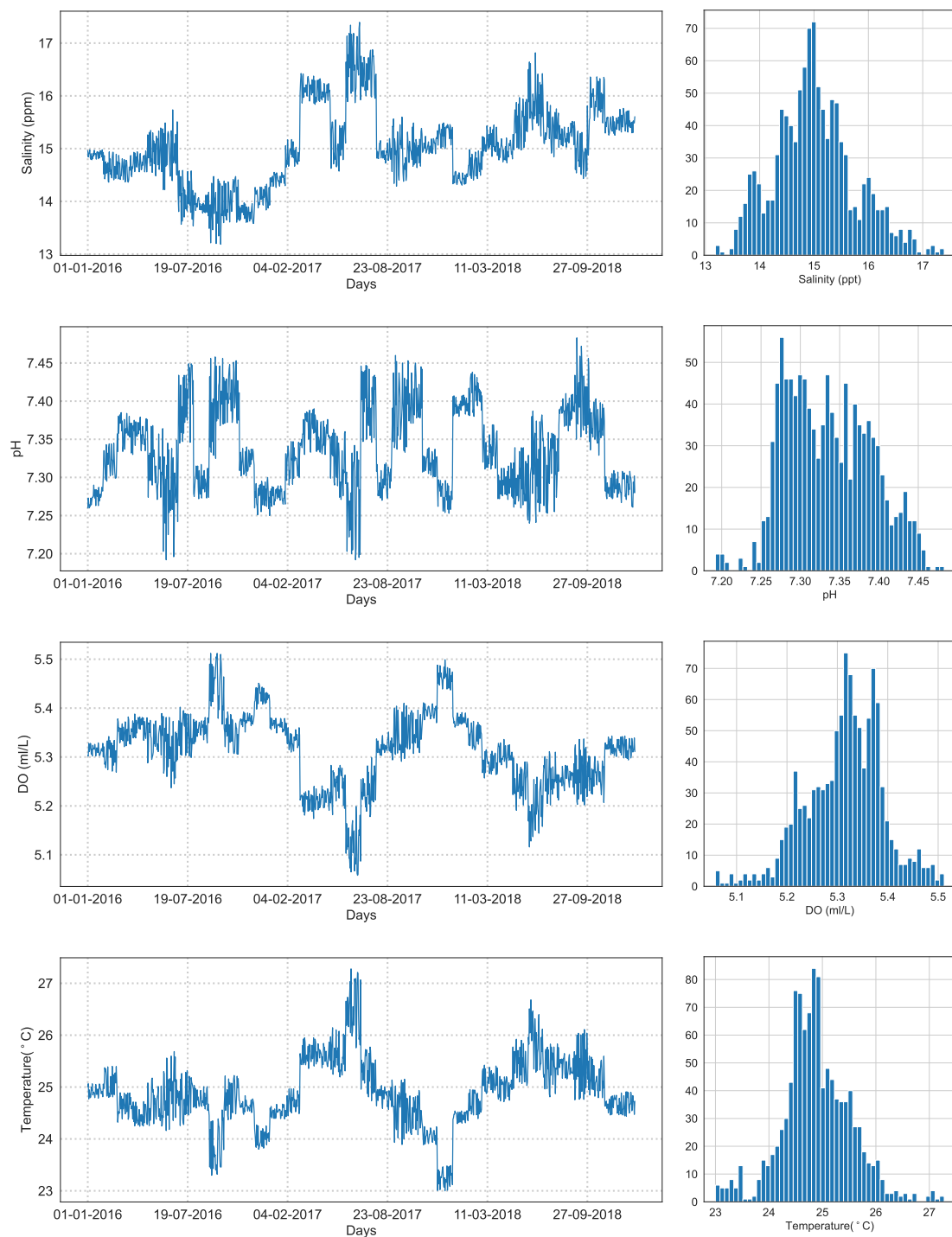


Figure 4.1: Water quality parameters variation trend of ADAK water quality dataset

4.2.2 Correlation analysis

Correlation analysis is performed on data from January 2016 and December 2018 to analyse the relationship between these WQ parameters (salinity, pH, DO and temperature). Pearson's correlation coefficient shows the correlation between the variables. A correlation ratio of more than 0.5 indicates a direct correlation with the parameter and a significant interdependence between the parameters. A value between 0.5 to 0.2 demonstrates a correlation with interdependence between them to some extent. A value of less than 0.1 suggests a poor correlation with the parameter and an insignificant interdependence between the parameters. The data is standardised to eliminate dimensional influence, and then the Pearson correlation coefficient matrix is applied. Table 4.2 shows the correlation coefficient matrix. The correlation between salinity, pH, DO, and temperature is given in Table 4.2. The findings in Table 4.2 show that temperature and salinity have a significant influence on the DO. Moreover, pH does not have any impact on the other WQ parameters monitored here.

Table 4.2: Correlation coefficient matrix of ADAK water quality dataset

	Salinity	pH	DO	Temperature
Salinity	1.0000	-0.0880	-0.7203	0.5278
pH	-0.0880	1.0000	0.0083	0.0185
DO	-0.7203	0.0083	1.0000	-0.9692
Temperature	0.5278	0.0185	-0.9692	1.0000

Temperature and DO show the most correlation. Whereas temperature and pH show the minimum correlation. The pH has an insignificant correlation with salinity, DO and temperature. DO has a higher correlation with temperature, followed by salinity and an insignificant correlation with pH. Similarly, the temperature has a high correlation with DO and a moderate correlation with salinity. However, the correlation of temperature with pH is insignificant.

4.2.3 Data pre-processing

Aquaculture WQ parameters data measured from the ponds may have anomalies such as missing data or abnormal data. This can be due to problems with sensors or possible errors while storing the data. These variations will lead to excessive deviation of predicted values from actual monitored values. In order to improve the accuracy of prediction, the predictive models must be provided with clean, reliable and succinct data. Unavailable data can no longer be retrieved, and missing data can only be computed as accurately as possible. Throughout this work, the missing part of the WQ parameters data is first filled using LIN algorithm [140], [141], and then this data is used to forecast the WQ parameters. As illustrated in Fig. 4.1, the WQ parameters

show continuous seasonal variation and also exhibit correlation with time.

Any data taken at definite time intervals forms time-series data. The dataset of WQ parameters acquired from a source also constitutes time-series data. It is an organized collection comprising measured values of WQ parameters at regular intervals. WQ parameters are measured at specific times every day.

$$TS_{i,n} = ((y_{i,1}, T_1), \dots, (y_{i,n}, T_n)) \quad (4.1)$$

In Eq. (4.1) $TS_{i,n}$ is defined as n -length time-series with i th WQ parameter. The sampling interval for all parameters is one day i.e., all parameters are measured at the same time on a daily basis. If the value $y_{i,v}$ at T_v is missing, then approximated value can be obtained using the LIN algorithm. The LIN function can be constructed as:

$$Y_{i,v} = y_{i,u} + \frac{y_{i,u} - y_{i,w}}{T_u - T_w}(T_v - T_u) \quad (4.2)$$

When a WQ data is missing at any point, the LIN algorithm first finds the two closest moments represented as T_u and T_w . Then it calculate the missing value at that moment T_v by utilizing the parameter values $y_{i,u}$ and $y_{i,w}$ at the moments T_u and T_w respectively based on Eq. (4.2). Here $Y_{i,v}$ represents the estimated value of missing value $y_{i,v}$.

4.3 DEEP LEARNING MODELS

4.3.1 LSTM DL-RNN

LSTM DL-RNN was proposed in 1997 by authors of [138] to avoid long-term dependency problems in RNN. In LSTM, a control unit is introduced to store information, unlike a hidden layer in RNN. This hidden state is divided to memory cells c_t and working memory h_t . The c_t is responsible for sequence features retention, and previous sequence memory is controlled by the forgetting gate f . The portion of the current memory c_t is controlled by output gate o_t and h_t is used as the output. The current state h_{t-1} and current input x_t written to memory cells are responsibility of the input gate i . The parameters associated with the LSTM model are collectively denoted as θ_{LSTM} , representing the set of weights and biases. The LSTM architecture [142] is shown in Fig. 4.2, and memory units are defined as follows in Eq. (4.3):

$$\begin{aligned}
h_t^{LSTM} &= H^{(L)}(h_{t-1}^{LSTM}, x_t, \theta_{LSTM}) \\
&= \begin{cases} f_t = \sigma(W_f \times [h_{t-1}, x_t] + b_f) \\ i_t = \sigma(W_i \times [h_{t-1}, x_t] + b_i) \\ \tilde{c}_t = \tanh(W_c \times [h_{t-1}, x_t] + b_c) \\ c_t = f_t \times c_{t-1} + i_t \times \tilde{c}_t \\ o_t = \sigma(W_o \times [h_{t-1}, x_t] + b_o) \\ h_t^{LSTM} = o_t \times \tanh(c_t) \end{cases} \quad (4.3)
\end{aligned}$$

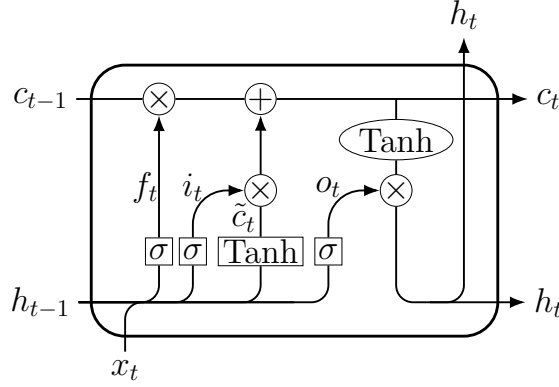


Figure 4.2: Forward LSTM cell structure

In this architecture f_t , i_t , and o_t are forget, input and output gate layer respectively. $\tilde{c}_t$ and c_t are new and final memory cell, w is weight matrices, b is bias vectors, σ is the sigmoid activation function.

4.3.2 GRU DL-RNN

GRU DL-RNN was proposed by authors of [139] in 2014. It similar to LSTM, but required less computing power. GRU is an improved version of RNN with only two gates, an update gate and a reset gate. There are no additional memory cells to store information; GRU can control information inside the unit. The update gate decides whether to pass the previous output h_{t-1} to the next cell. The reset gate reads the input sequences when the gate is set to zero and forgets the previously calculated state. As a result, GRU has fewer tensor operations than LSTM and runs typically faster than LSTM. The parameters associated with the GRU model are collectively denoted as θ_{GRU} , representing the set of weights and biases. The GRU architecture [142] is shown in Fig. 4.3 and memory units are defined as follows in Eq. (4.4):

$$h_t^{GRU} = H^{(G)}(h_{t-1}^{GRU}, x_t, \theta_{GRU})$$

$$= \begin{cases} z_t = \sigma(W_z \times [h_{t-1}, x_t]) \\ r_t = \sigma(W_r \times [h_{t-1}, x_t]) \\ \tilde{h}_t = \tanh(w \times [r_t \times h_{t-1}, x_t]) \\ h_t^{GRU} = (1 - z_t) \times h_{t-1} + z_t \times \tilde{h}_t \end{cases} \quad (4.4)$$

Figure 4.3: Forward GRU cell structure

In this architecture z_t is the reset and r_t is the update gate, $\tilde{h}_t$ is process input, and h_t is hidden state update, w is weight matrices and σ is the sigmoid activation function.

4.3.3 Bidirectional LSTM/GRU DL-RNN

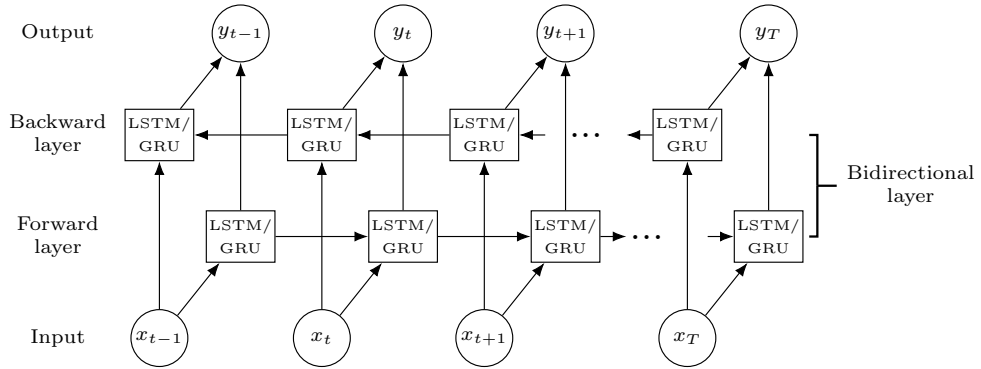


Figure 4.4: Structure of Bi-LSTM/Bi-GRU

The bidirectional DL models consist of two processes in both forward and backward directions, as depicted in Fig. 4.4. The processing in both directions helps the model capture patterns that the unidirectional LSTM and GRU models cannot capture. The bi-directional DL-RNNs are an extension of LSTM and GRU. Bi-directional DL-RNNs can fit data in both forward and backward directions and concatenate the prediction, while LSTM/GRU can only fit data in one direction [143]. A Bi-LSTM/Bi-GRU model first feeds data into one of the LSTM/GRU models for training the given data sequence, i.e., feedback layer and re-training with another LSTM/GRU model, but

with the opposite order of the input data sequence. Bi-LSTM/Bi-GRU model can capture patterns ignored by LSTM/GRU because it has added a reverse directional layer. Double LSTM/GRU application improves long-term study and thereby improves accuracy [144]. Bi-directional networks are the extension of the LSTM/GRU model, which can learn about the relation of past and future factors of load and current load, which helps in extracting the in-depth features of loaded data [145]. The bi-directional LSTM/GRU structure is shown in Fig. 4.4.

4.3.4 Model implementation

The WQ dataset is used to train the DL models to predict each WQ parameter. First, the dataset is divided into two; 80% for the training and 20% for the testing. The training dataset is used to develop the models, while the testing dataset is used to validate and compare the performance of models developed. All input and output variables were scaled between 0 and 1 via normalization and minimum-maximum scaling techniques, using the MinMaxScaler in the scikit-learn preprocessing library using Python [146]. Moreover, the Adam optimizer is employed for training the models due to its adaptive learning rate mechanism and ability to handle sparse gradients effectively [147]. The training and testing datasets are transformed into supervised learning frames with various array manipulation techniques. The time series sequences are converted to input-output equations using the sliding window procedure. This transformed data is used to train the models for WQP.

4.4 RESULTS AND DISCUSSIONS

The objective is to analyse the performance of DL models, LSTM, GRU, Bi-LSTM, and Bi-GRU to predict aquaculture WQ parameters (salinity, ph, DO and temperature). In this section, the performance of DL models is studied with a fixed set of h_p on the two datasets.

4.4.1 ADAK water quality dataset

Figure 4.5 compares the predicted values with the true data on the ADAK WQ dataset using different DL models. All these models are trained using 80% of the ADAK WQ dataset and tested with 20% of the data. The predicted values are compared with the true data. Here a fixed set of h_p (epochs = 100, learning rate = 0.0008, batch size = 32, window size = 30 and Adam optimizer) is used. The models are evaluated using performance metrics MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each of the models is also calculated.

The performance of standalone DL models, such as LSTM, GRU, Bi-LSTM, and

Table 4.3: Prediction accuracy of LSTM, GRU, Bi-LSTM and Bi-GRU for ADAK water quality dataset

Water Quality Parameter	Model	Training Period					Testing Period				
		MAE	MSE	RMSE	MAPE	Time (s)	MAE	MSE	RMSE	MAPE	Time (s)
Salinity (ppt)	LSTM	0.2574	0.1135	0.3368	0.0172	25.44	0.2388	0.0939	0.3065	0.0154	0.06
	GRU	0.2315	0.0952	0.3085	0.0155	26.65	0.2436	0.1019	0.3191	0.0159	0.06
	BiLSTM	0.2486	0.1089	0.3300	0.0165	33.66	0.2877	0.1268	0.3561	0.0185	0.06
	BiGRU	0.2231	0.0903	0.3005	0.0149	35.36	0.2309	0.0874	0.2956	0.0149	0.06
pH	LSTM	0.0249	0.0012	0.0339	0.0034	27.81	0.0239	0.0010	0.0314	0.0033	0.05
	GRU	0.0302	0.0015	0.0391	0.0041	28.30	0.0311	0.0015	0.0391	0.0042	0.06
	BiLSTM	0.0244	0.0011	0.0331	0.0033	34.12	0.0239	0.0010	0.0319	0.0033	0.06
	BiGRU	0.0327	0.0017	0.0408	0.0044	35.15	0.0330	0.0017	0.0406	0.0045	0.06
DO (ml/L)	LSTM	0.0333	0.0018	0.0420	0.0063	26.67	0.0305	0.0014	0.0373	0.0058	0.06
	GRU	0.0255	0.0011	0.0338	0.0048	27.00	0.0263	0.0012	0.0348	0.0050	0.06
	BiLSTM	0.0235	0.0010	0.0309	0.0044	34.93	0.0246	0.0010	0.0313	0.0047	0.06
	BiGRU	0.0263	0.0012	0.0340	0.0050	38.10	0.0252	0.0011	0.0325	0.0048	0.06
Temperature(°C)	LSTM	0.3030	0.1495	0.3866	0.0123	26.69	0.3275	0.1748	0.4181	0.0131	0.05
	GRU	0.2822	0.1325	0.3640	0.0113	28.13	0.2784	0.1318	0.3630	0.0110	0.06
	BiLSTM	0.2582	0.1225	0.3501	0.0104	36.69	0.2469	0.1012	0.3181	0.0098	0.07
	BiGRU	0.3269	0.1806	0.4249	0.0131	36.29	0.3093	0.1535	0.3918	0.0122	0.09

$h_p : L_r = 0.0008, E_p = 100, W_z = 30, B_z = 32$, and Adam optimizer.

Table 4.4: Prediction accuracy of LSTM, GRU, Bi-LSTM and Bi-GRU for MAC water quality dataset

Water Quality Parameter	Model	Training Period					Testing Period				
		MAE	MSE	RMSE	MAPE	Time (s)	MAE	MSE	RMSE	MAPE	Time (s)
Salinity (ppt)	LSTM	0.1216	0.0198	0.1407	0.0035	131.11	0.2590	0.9396	0.9693	0.0078	0.44
	GRU	0.0156	0.0005	0.0217	0.0005	135.46	0.0905	0.5058	0.7112	0.0029	0.45
	BiLSTM	0.0224	0.0011	0.0338	0.0007	167.55	0.0394	0.0121	0.1101	0.0012	0.52
	BiGRU	0.0453	0.0024	0.0489	0.0013	180.59	0.1234	0.3936	0.6274	0.0038	0.47
pH	LSTM	0.0078	0.0001	0.0090	0.0009	188.08	0.0093	0.0001	0.0111	0.0011	1.03
	GRU	0.0521	0.0030	0.0550	0.0063	203.74	0.0570	0.0077	0.0878	0.0067	0.63
	BiLSTM	0.0071	0.0001	0.0092	0.0009	226.54	0.0118	0.0004	0.0198	0.0014	0.70
	BiGRU	0.0266	0.0007	0.0270	0.0032	225.32	0.0582	0.0152	0.1233	0.0067	0.59
DO (ml/L)	LSTM	0.0648	0.0081	0.0898	0.0204	166.33	0.1114	0.0527	0.2295	0.0209	0.56
	GRU	0.1213	0.0224	0.1495	0.0380	131.55	0.2427	0.6086	0.7801	0.0391	0.38
	BiLSTM	0.0792	0.0103	0.1014	0.0206	170.38	0.1096	0.0295	0.1717	0.0259	0.52
	BiGRU	0.0981	0.0151	0.1231	0.0241	173.14	0.4637	2.1555	1.4682	0.0534	0.47
Temperature(°C)	LSTM	0.6193	0.3871	0.6222	0.0246	138.20	0.6209	0.3955	0.6289	0.0227	0.45
	GRU	0.6364	0.4208	0.6487	0.0252	134.23	0.7345	0.5432	0.7370	0.0268	0.40
	BiLSTM	0.1039	0.0129	0.1136	0.0040	173.03	0.1413	0.0212	0.1454	0.0052	0.51
	BiGRU	0.3063	0.0953	0.3086	0.0121	174.85	0.3437	0.1196	0.3459	0.0125	0.57

$h_p : L_r = 0.0008, E_p = 100, W_z = 30, B_z = 32$, and Adam optimizer.

Bi-GRU are evaluated for WQP. Figure 4.5 and Fig. 4.6 illustrate the comparative analysis of predicted values versus actual data from the ADAK and MAC WQ datasets, respectively, using these DL models.

4.4.2 MAC water quality dataset

Figure 4.6 compares the prediction of all DL models with the true data on the MAC WQ dataset. All these models are trained using 80% of the ADAK WQ dataset and predicted the 20% values. The predictions are compared with the true data. Here a fixed set of h_p (epochs = 50, learning rate = 0.0008, batch size = 64, window size = 30 and Adam optimizer) is used. The models are evaluated using MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each

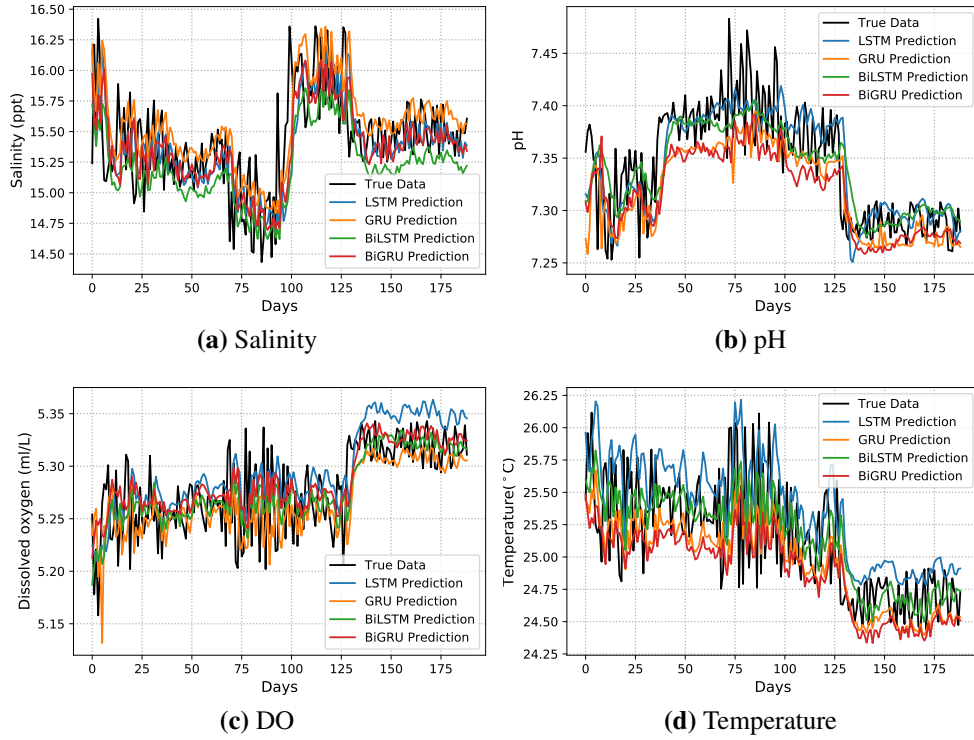


Figure 4.5: Comparison of predicted values using LSTM, GRU, Bi-LSTM, and Bi-GRU predictions and ADAK water quality observed data. (a) Salinity, (b) pH, (c) DO, (d) Temperature

of the models is also calculated.

As demonstrated in Table 4.3 and Table 4.4, these models have shown varying degrees of performance in predicting different parameters of the ADAK and MAC WQ datasets. All models exhibit a level of competence, yet distinct strengths and limitations are observed. The LSTM model, for example, offers a satisfactory performance across both datasets, although it tends to be slightly less accurate compared to the BiLSTM and GRU models in certain parameters. The BiLSTM model, despite its bidirectional learning mechanism, does not consistently outperform its unidirectional counterparts (LSTM and GRU), suggesting that the additional complexity may not always translate to superior performance.

It's important to note that while these models show adequate performance, none of them significantly outperforms the others in predicting all parameters for distinct datasets. This indicates the need for more sophisticated models to better predict WQ parameters, and serves as a prelude to the introduction of hybrid models in the next chapter, which aim to combine the strengths of these individual models for superior predictive performance.

Table 4.5 provides a comprehensive performance comparison of DL-RNN models, encompassing LSTM, GRU, Bi-LSTM, and Bi-GRU, provides an insightful

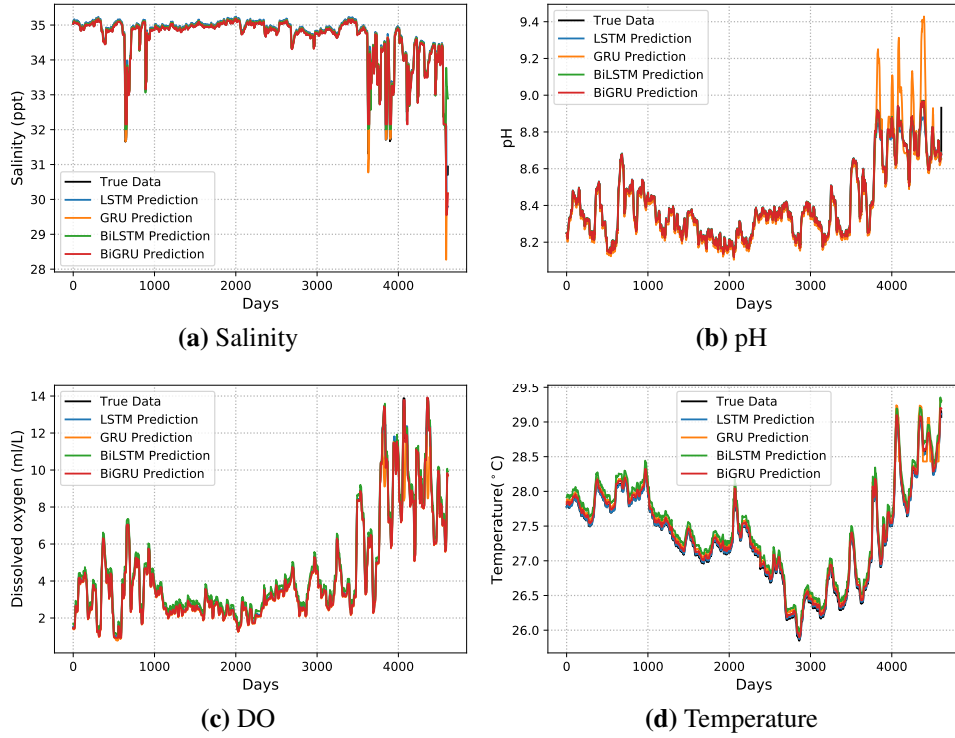


Figure 4.6: Comparison of predicted values using LSTM, GRU, Bi-LSTM, and BiGRU predictions and MAC water quality observed data. (a) Salinity, (b) pH, (c) DO, (d) Temperature

Table 4.5: Comprehensive Performance Comparison of DL-RNN Models

Model	ADAK Dataset					MAC Dataset				
	MAE	MSE	RMSE	MAPE	Time (s)	MAE	MSE	RMSE	MAPE	Time (s)
LSTM	0.1547	0.0665	0.1998	0.0098	26.65	0.2034	0.1038	0.2154	0.0124	155.93
GRU	0.1424	0.0576	0.1864	0.0089	27.52	0.2064	0.1117	0.2187	0.0175	151.25
Bi-LSTM	0.1387	0.0584	0.1860	0.0087	34.85	0.0532	0.0061	0.0645	0.0066	184.38
Bi-GRU	0.1523	0.0685	0.2001	0.0094	36.22	0.1191	0.0284	0.1269	0.0102	188.48

Values represent the average of the respective metric across all parameters.

overview of their effectiveness in predicting water quality parameters across two distinct datasets: ADAK and MAC. The evaluation metrics employed, including MAE, MSE, RMSE, MAPE values, and computation Time, offer a holistic view of each model's performance.

4.5 SUMMARY

In this chapter, the focus is on the application of DL-RNN models with Adam optimization for aquaculture WQP. The chapter explores the use of DL-RNN models, including LSTM, GRU, and their bidirectional counterparts for WQP. The main objectives are to compare the performance of these DL models and identify the most accurate and efficient one for the task at hand. The ADAK and MAC datasets are used for experimental results and comparisons. The chapter provides valuable insights into the performance of various DL-RNN models for aquaculture WQP. While all models

show competence in predicting WQ parameters, no single model significantly outperforms the others in all cases. This suggests the need for more sophisticated approaches, leading to the introduction of hybrid models in subsequent chapters.